SYNAPSE + HORIZON EDGE PROTECTION PLATFORM

Platform Architecture

End-to-end view of the Horizon edge protection platform. Sensors at the edge, a central hub in the middle, storage and consumers at the back. Every edge decision is local; every signal flows to the hub asynchronously; the hub stays out of the request path.

∞	<100ms	5-tier	0
SENSORS PER FLEET	SIGNAL SYNC	ARCHITECTURE	EDGE DEPS

LAYERED ARCHITECTURE

Top to bottom: request origin → edge decision → hub intelligence → persistence → consumers

01 CLIENTS — REQUEST ORIGIN

- **Browser** — Human traffic
- **Mobile App** — Native clients
- **API Client** — Machine-to-machine
- **Crawler** — Good bots · verified DNS
- **Attacker** — Bots · scanners · botnets

▼ HTTPS — request + headers + body

02 EDGE · SYNAPSE SENSORS — <1ms LOCAL DECISION

- **Synapse US-East** — Pingora proxy + WAF + DLP + Bot
- **Synapse EU-West** — Pingora proxy + WAF + DLP + Bot
- **Synapse AP-South** — Pingora proxy + WAF + DLP + Bot

237 rules · 500+ bot signatures · 22+ DLP patterns · 8 correlation detectors · JA4/JA4H · Actor cache 100K · Auto-mitigation

▼ signals · ▲ commands

03 CONTROL PLANE · HORIZON HUB — CENTRALIZED INTELLIGENCE

- **WS Gateway** — Signal ingest · fan-in
- **Aggregator** — Dedupe · batch · persist
- **Correlator** — Cross-sensor campaigns
- **Broadcaster** — Real-time push to UI
- **REST API** — 80+ endpoints · admin

Fleet manager · Rule distributor · Sigma engine · CyberChef · War room · Hunt API · Remote shell tunnels

▼ persist · ▲ query

04 STORAGE — STATE & ANALYTICS

- **PostgreSQL** — Source of truth · tenants · rules · fleet state
- **ClickHouse** — Time-series · signal history · hunt queries
- **Redis** — Cache · pub/sub · multi-instance sessions

▲ read · realtime push

05 CONSUMERS — OPERATORS & INTEGRATIONS

- **Horizon UI** — SOC dashboard · live map
- **Synapse CLI** — Admin · rules · diagnostics
- **Webhooks** — Slack · PagerDuty · SIEM
- **Apparatus Lib** — Embedded SDK · drills

COMPONENT RESPONSIBILITIES

COMPONENT	LAYER	RESPONSIBILITY
Synapse	EDGE Rust · Pingora	Inline WAF, DLP, bot detection, behavioral profiling, interrogation. Makes every allow/block decision locally — no network calls.
WS Gateway	HUB Node.js	Accepts persistent WebSocket connections from sensors, authenticates each tenant, streams signals into the aggregator.
Aggregator	HUB Node.js	Deduplicates, batches, and writes signals. Dual-write to PostgreSQL (hot) and ClickHouse (cold). Failure of either doesn't block ingest.
Correlator	HUB Node.js	Cross-sensor campaign correlation. Links distributed attacks via JA4 fingerprints, auth tokens, ASN proximity. Emits campaign events.
Broadcaster	HUB Node.js	Pub/sub fan-out from aggregator and correlator to Horizon UI. Real-time threat map updates, campaign timelines, SOC alerts.
REST API	HUB Fastify	80+ endpoints for admin, hunt queries, rule management, drill orchestration, and external integrations. Rate-limited per tenant.
PostgreSQL	STORE primary	Tenants, rules, config, fleet state, recent signals. Row-level tenant isolation. The source of truth for everything operational.
ClickHouse	STORE analytics	Full signal history for time-window queries, hunt jobs, and aggregation. Write-heavy, read-optimized for SOC workflows.
Redis	STORE cache	Session sharing across Horizon instances, rate limit counters, correlation pub/sub, and multi-instance coordination.

RESILIENCE MODEL

The hub is additive intelligence — not a dependency

HUB ONLINE — FULL FLEET MODE

Centralized intelligence & cross-fleet correlation

- Campaign correlation across every sensor in the fleet
- Real-time rule distribution — ~240µs hot reload per sensor
- Shared blocklists and IoC propagation across tenants
- Live threat map, hunt queries, and SOC war-room workflows
- Anonymized cross-tenant signal sharing via SHA-256 fingerprints

HUB OFFLINE — INDEPENDENT EDGE MODE

Every sensor continues protecting its edge

- All WAF, DLP, and bot decisions keep running locally
- Per-sensor campaign correlation and actor scoring unaffected
- Signals buffer locally and replay when the hub returns
- Last-known rules, blocklists, and challenge state remain in effect
- No blocked requests, no dropped traffic, no customer impact

END-TO-END TIMING

OPERATION	LATENCY	PATH	TARGET
Request → Edge decision	<1ms	Client → Synapse → Origin	✓ Met
Signal → Hub ingest	<100ms	Synapse → WS Gateway → Aggregator	✓ Met
Campaign correlation window	~1s	Aggregator → Correlator → Broadcaster	✓ Met
Rule push → sensor applied	~240µs	Horizon API → WS command → Synapse reload	✓ Met
UI update (live threat map)	<500ms	Broadcaster → UI WebSocket → render	✓ Met

Independent at the edge. Intelligent at the hub. Resilient by design.

Every sensor decides locally; every decision flows to the fleet. The hub never gates a request — it amplifies every one that happens.